

Demetrius Hernandez

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EDUCATION

The University of Notre Dame

Ph.D. in Computer Science

Notre Dame, IN

Expected: Dec. 2028

- Advisor: Dr. Jane Cleland-Huang
- Research Focus: Decision-making and reasoning for autonomous drones in emergency response scenarios

The University of Texas at El Paso (UTEP)

B.S. in Computer Science, Minor in Mathematics (Cum Laude)

El Paso, TX

Dec. 2022

- Coursework in Research Methods, Machine Learning, Deep Learning, Data Structures, Data Mining, Database Management, Discrete Mathematics, Probability and Statistics, Matrix Algebra

TECHNICAL SKILLS

Proficient Programming Languages: Python, R, Julia, C++, MATLAB

Machine Learning & AI: TensorFlow, Keras, PyTorch, Scikit-learn, Reinforcement Learning, Deep Learning

Other Tools & Frameworks: Docker, Kubernetes, Git, .NET, Linux

WORK EXPERIENCE

PhD Researcher

The University of Notre Dame

Aug. 2024 – PRESENT

Notre Dame, IN

- National Science Foundation (NSF) Research Fellow
- Autonomous drones for emergency response scenarios

Computer Scientist

White Sands Missile Range, Counter Drone Team

Feb. 2023 – Aug. 2024

White Sands Missile Range, NM

- Awarded FY 2023 DOD SMART Scholar of the Year Award
- Awarded the Commanders Coin for Excellence by the White Sands Test Center Commander
- Spearheaded automation of analysis processes, conducted research on Graph Theoretic Approaches for Evaluation of Counter Drone Systems, and developed a neural network-based simulation framework, reducing testing costs.

Undergraduate Researcher

UTEP Computer Science Department

Jan. 2022 – Dec. 2022

El Paso, TX

- Collaborated with interdisciplinary teams to enhance long-read alignments of DNA/RNA sequences and built neural network architectures, utilizing TensorFlow, to improve DNA assembly through encoding existing minimizer schemes.

Undergraduate Research Assistant

UTEP Brain Computation Lab

Sep. 2021 – Jan. 2022

El Paso, TX

- Researched and implemented a comprehensive decision-making model emulating brain circuits, contributing to the lab's objective of integrating big data analysis and behavioral physiology to investigate psychiatric and neurological disorders

Software Engineering Intern

White Sands Missile Range, Survivability and Vulnerability Directorate

May 2022 – Aug. 2022

White Sands Missile Range, NM

- Built software to control instrumentation and collect data from equipment such as antennas and spectrum analyzers. Ensured compliance with electromagnetic (EM) requirements for testing weapons systems in EM environments.

Research Experience for Undergraduates

Oregon State University

Jul. 2021 – Sep. 2021

Corvallis, OR

- Utilized machine learning techniques to conduct research in quantitative ecology, analyzing community science data and adapting occupancy modeling frameworks to emerging datasets from the eBird app.

Undergraduate Research Assistant

UTEP Cyber-Share Center of Excellence

Jul. 2020 - Jul. 2021

El Paso, TX

- Analyzed and visualized the Cafeteria Roenbergensis virus, employing high-performance computing resources to contribute to the understanding of its intricate structure composed of approximately 120 million atoms.

Disney College Program Intern

Aug. 2019 – Jan. 2020

The Walt Disney Company

Lake Buena Vista, FL

- Participated in Disney's Ultimate EnginEARing Exploration, Engineering Exploration seminars, and Disney Hackathon, developing transferable skills in problem-solving, teamwork, guest service, and effective communication.

PUBLICATIONS

Graph Theoretic Approaches for Evaluating Counter Drone Systems, Hernandez, D. *17th NATO Operations Research and Analysis Conference Proceedings, 2023.*

Aquatic Ecotoxicology: Theoretical Explanation of Empirical Formulas, Hernandez D., Holguin G., Parra F., Sanchez V., Kreinovich V. *Uncertainty, Constraints, and Decision Making. Studies in Systems, Decision and Control, vol 484.*

FELLOWSHIPS AND GRANTS

(2025) Notre Dame-IBM Technology Ethics Lab Call for Proposals: Co-author and awarded \$38,000 to support research on human-drone collaboration and AI ethics.

(2024) National Science Foundation (NSF) CSGrad4US Fellowship: Awarded to persons in industry with demonstrated potential for doctoral research. \$37,000 for 3 years with an additional \$16,000 per year for cost-of-education.

(2023) Department of Defense (DoD) Creative Research and Engineering Advancing Technical Equity in STEM Grant: \$10,000 award for custom drone development in support of enhancing counter-drone testing capabilities.

(2022) RECOMB Travel Fellowship: RECOMB Travel fellowship

(2021) DOD SMART Scholarship: Full-tuition for undergrad and \$25,000/year stipend.

(2018) UTEP Andalusite Award: \$3,000 per year for 4 years.

AWARDS AND HONORS

(2025) Best Oral Presentation: Graduate Research Symposium, University of Notre Dame

(2025) Finalist: Three Minute Thesis (3MT) Competition, University of Notre Dame

(2025) Honorable Mention: Policy Pitch Competition, Notre Dame Science Policy Initiative

(2024) Congressional Recognition: Received a Certificate of Special Congressional Recognition

(2024) Commanders Coin for Excellence: Awarded by the Commander of White Sands Test Center

(2024) DOD SMART Scholar of the Year: Department of Defense SMART Scholarship Program

(2022) Best Poster Presentation: UTEP Undergraduate Research Symposium

(2021) First Place: Association for Computing Machinery (ACM) Sunrise Hackathon

TALKS

(2023) DOD Weapons System Software Summit: Machine Learning-Powered Software Safety Categorization Using Software Requirements Specifications (SRS) in DOD Weapon Systems

(2023) 17th NATO Operations Research and Analysis Conference: Graph Theoretic Approaches for Evaluating Counter Drone Systems

(2022) Joint UTEP/NMSU Workshop of Mathematics and Computer Science: Efficient Minimizer Schemes using Deep Networks

POSTER PRESENTATIONS

(2022) Great Minds in STEM Conference: Improving DNA Assembly with Machine Learning

(2022) 26th International Conference on Research in Computational Molecular Biology: Improving DNA Assembly with Machine Learning

LEADERSHIP AND SERVICE

Mentor

Aug. 2021 – PRESENT

Army Educational Outreach Gains in the Education of Mathematics and Science

White Sands Missile Range, NM

- Mentored high school students in the AEOP GEMS summer program, providing guidance, facilitating workshops, and fostering a positive learning environment, with a commitment to continuing in the role.